

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

Introduction

- Definition of FIR Digital Filters
- Characteristics of FIR Digital Filters
- Specifications of FIR Digital Filters
- Design of FIR Digital Filters

1. Definition of FIR Digital Filters

Discrete LTI System $H(z)=\dfrac{\sum_{j=0}^{M} b_j z^{-j}}{1+\sum_{i=1}^{N} a_i z^{-i}}$

If a_i=0

Finite Length Impulse Response

$$H(z)=\sum_{j=0}^{M} b_j z^{-j} \Rightarrow h[k]=\mathcal{Z}^{-1}\{H(z)\}=\{b_0, b_1, \cdots b_M\}$$

FIR （Finite Impulse Response） Digital Filters

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2. Characteristics of FIR Digital Filters

IIR $$H(z)=\frac{\sum_{j=0}^{M} b_j z^{-j}}{1+\sum_{i=1}^{N} a_i z^{-i}}$$

FIR $$H(z)=\sum_{j=0}^{M} b_j z^{-j}$$

※ The system may be unstable

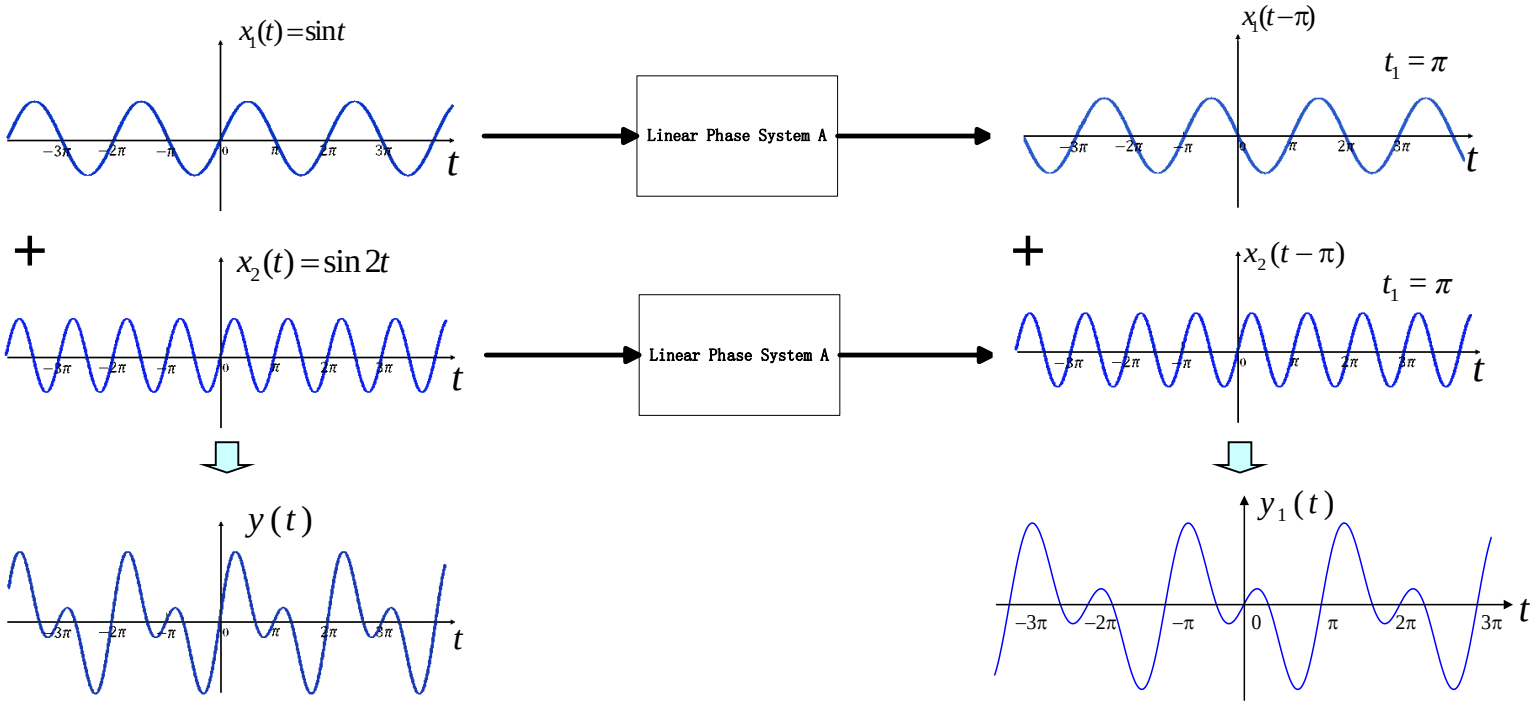
※ The system has non-linear phase

※ FIR system is always stable

※ The system can have a linear phase

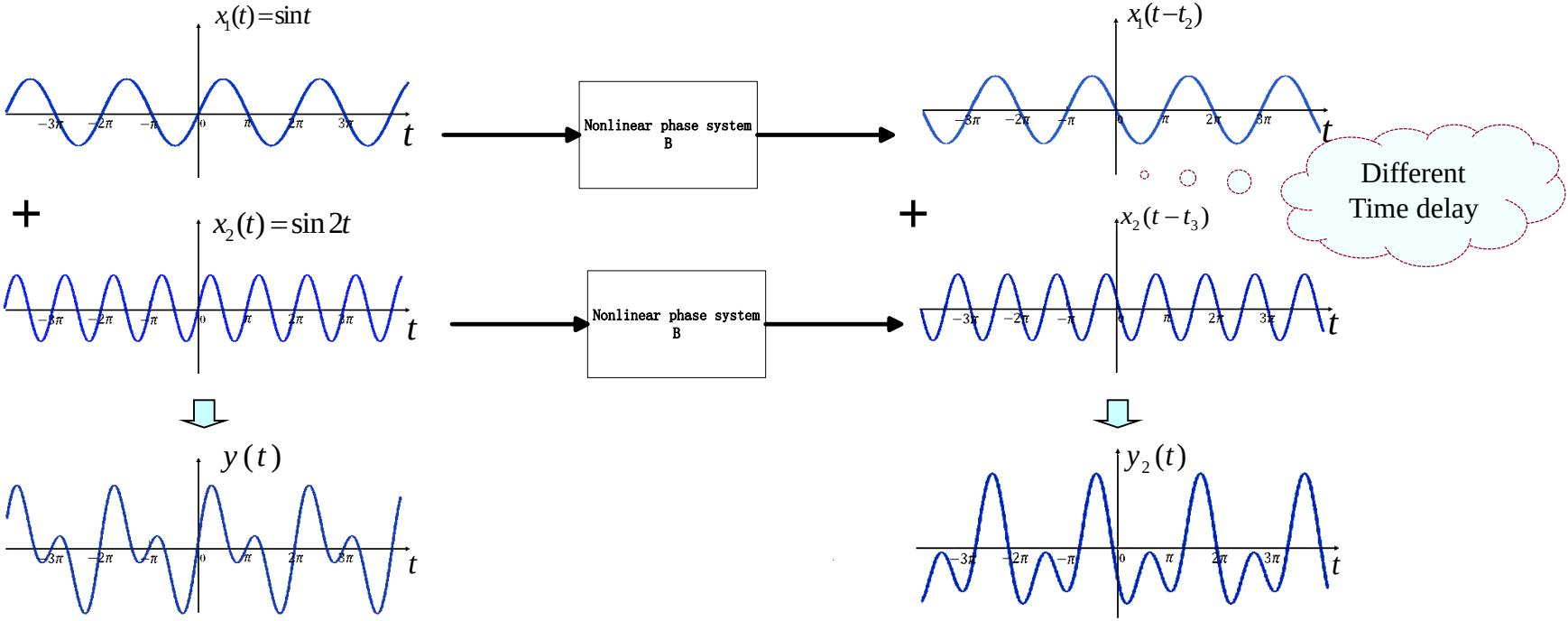
2. Characteristics of FIR Digital Filters

Linear phase



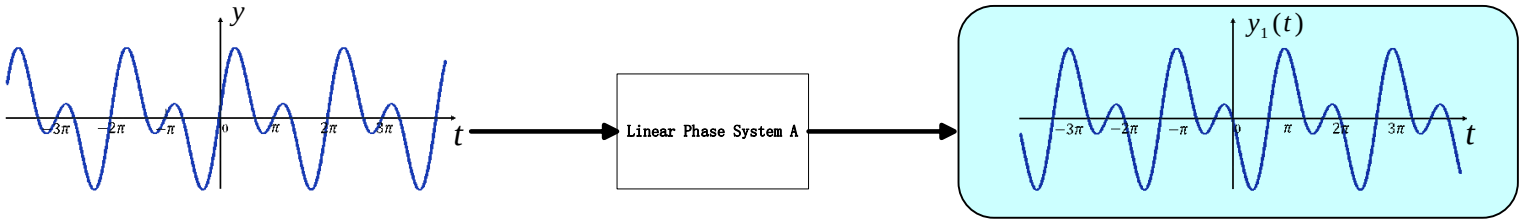
2. Characteristics of FIR Digital Filters

Linear phase

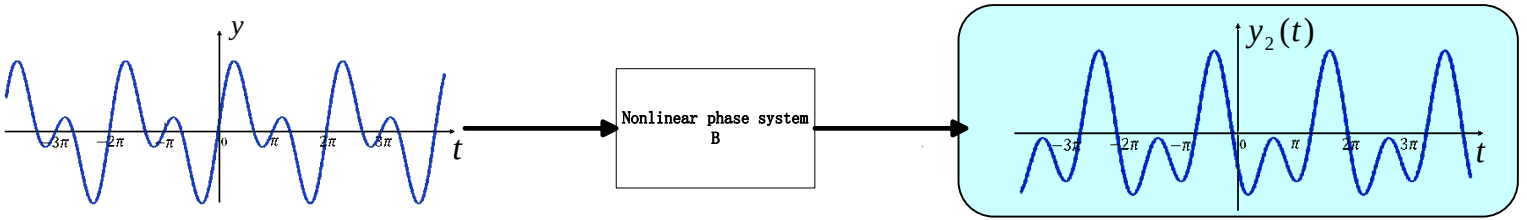


2. Characteristics of FIR Digital Filters

Linear phase

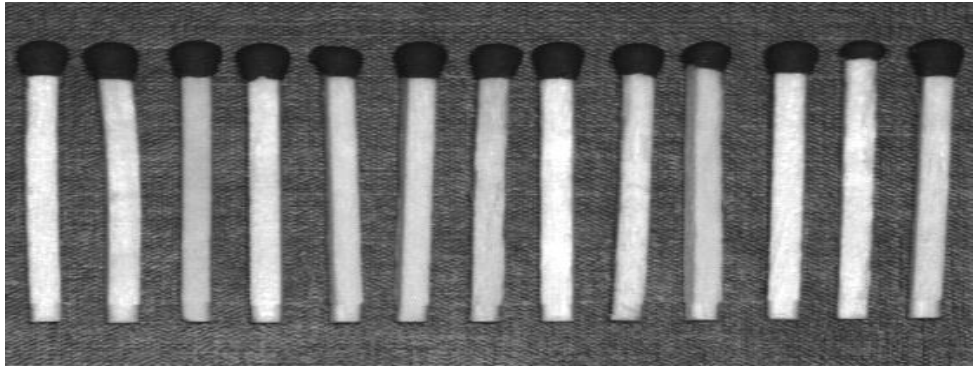


"Compared to the original superimposed signal, there is only a time delay."

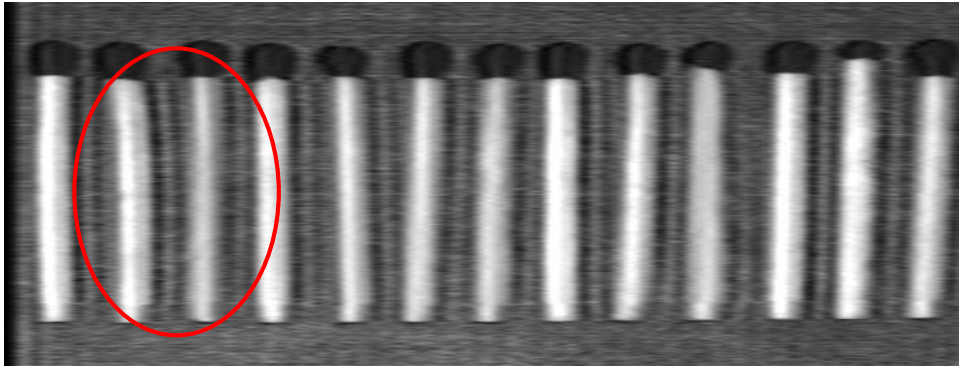


Signal distortion

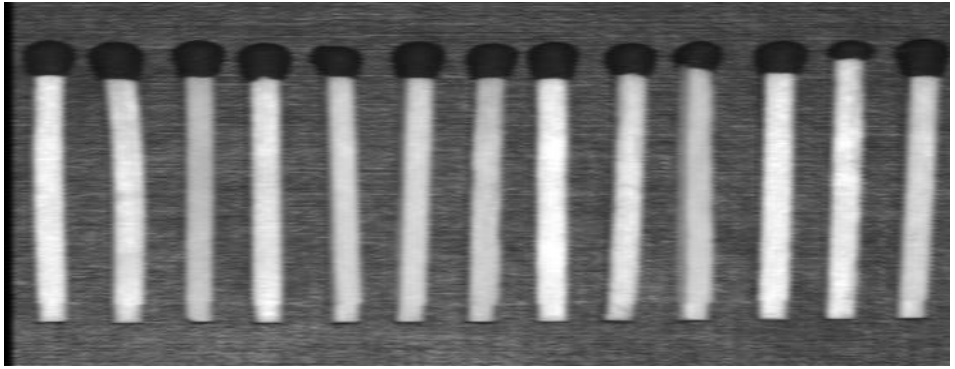
2. Characteristics of FIR Digital Filters



Original image



Passing through a non-linear phase system

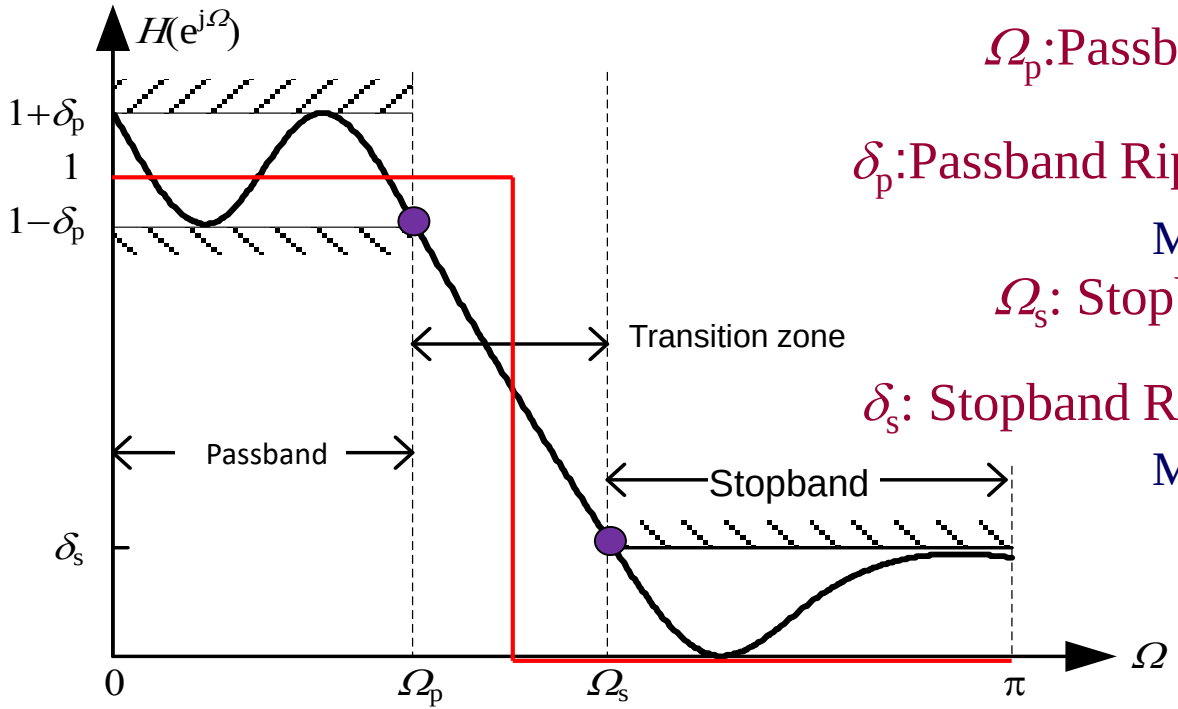


Passing through a linear phase system

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3. Specifications of FIR Digital Filters



Ω_p:Passband Edge Frequency

δ_p:Passband Ripple ⟹ $A_p = -20\lg(1-\delta_p)$

Maximum Passband Attenuation(dB)

Ω_s: Stopband Edge Frequency

δ_s: Stopband Ripple ⟹ $A_s = -20\lg\delta_s$

Minimum Stopband Attenuation(dB)

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4. Design of FIR Digital Filters

Design of Digital Filters: Determine the digital filter system function $H(z)$ based on the given design specifications.

For FIR digital filters, due to

$$H(z)=\sum_{k=0}^{M} b_k z^{-k} = \sum_{k=0}^{M} h[k] z^{-k}$$

Therefore, in designing FIR digital filters, it is only necessary to determine $h[k]$.

4. Design of FIR Digital Filters

Common Methods for FIR Filter Design:

※ Windowing Method

※ Frequency Sampling Method

※ Optimal Design Method

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.